

Probabilistic Reasoning Over Time

1 Markov Chains

1.1 Definitions

Stochastic Process A stochastic process is a family of random variables $\{X(t): t \in T\}$ defined on a *probability space*, where T is the *parameter set*. When $T = \{0, 1, 2, 3, \dots\}$, it is usually referred to as a time series. When T is a vector space, the stochastic process is called a random field. The collection of all possible states (values) of $X(t)$ forms the *state space*, which can be either discrete or continuous.

Markov Assumption / Markov Property; Markov Process / Markov Chain A Markov process (or Markov chain) is a stochastic process indexed by time, which satisfies the Markov assumption (or Markov property) — namely, the *conditional probability distribution* of the current state depends on only a *finite fixed number* of previous states, i.e.,

$$P(X_t | \{X_{t-1}, X_{t-2}, \dots\}) = P(X_t | \{X_{t-1}, X_{t-2}, \dots, X_{t-k}\}) \quad \text{for any } t$$

First-Order Markov Process A first-order Markov process is a Markov process where the current state depends only on the immediately preceding state, i.e.,

$$P(X_t | \{X_{t-1}, X_{t-2}, \dots\}) = P(X_t | X_{t-1}) \quad \text{for any } t$$

In the following sections, we will focus on first-order Markov processes. Without specification, a Markov process is assumed to be first-order.

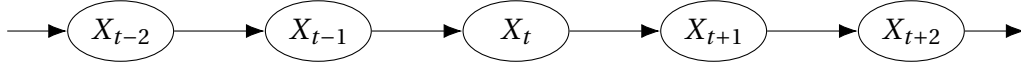


Figure 1: A first-order Markov process.

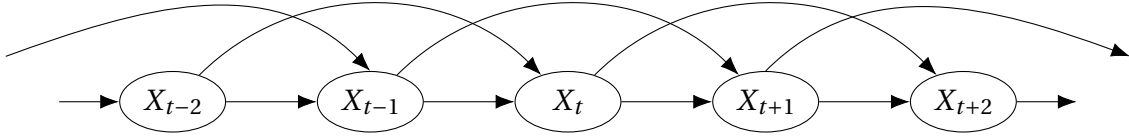


Figure 2: A second-order Markov process.

Transition Model In a *first-order* Markov process, the transition model is the conditional probability distribution $P(X_t | X_{t-1})$. A transition model can be represented in several ways. The following figure shows different representations of the same transition model for a first-order Markov process.

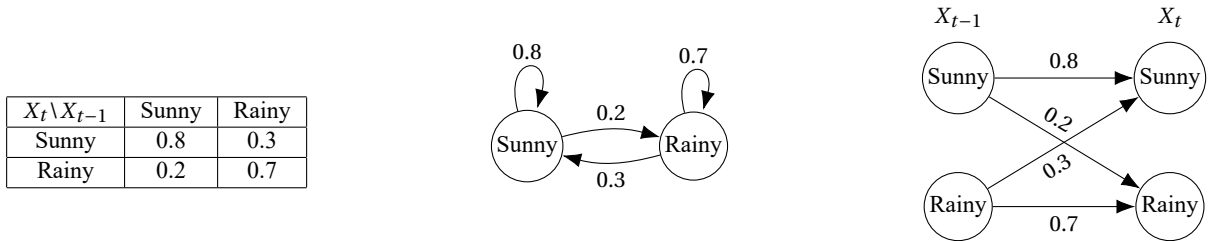


Figure 3: A transition model in three different representations.

From above, we can see that Markov processes are a special class of *Bayesian networks*. And the transition model is the *conditional probability table* (CPT) of the first-order Markov process. Compared to a general Bayesian network, a first-order Markov process has a simpler structure (a chain) and fewer parameters (only one CPT).

1.2 Markov Chain Inference

Inference on Markov chains is basically the same as inference in Bayesian networks.

Forward Algorithm Forward algorithm is a dynamic programming algorithm used to solve queries about the current state given a previous state. It is simply based on the following equation:

$$P(X_t) = \sum_{x \in S} P(X_t | X_{t-1} = x) P(X_{t-1} = x) \quad \text{where } S = \{\text{all possible states}\}$$

Matrix representation The forward algorithm can be expressed in matrix form. Suppose the set of all possible states is given by $S = \{s_1, s_2, \dots, s_n\}$. Then we can represent the transition model as an $n \times n$ matrix $\mathbf{M}$, where the entry $\mathbf{M}[i, j] := P(X_t = s_i | X_{t-1} = s_j)$. Let the vector $\mathbf{p}_t$ represent the probability distribution of X_t , i.e., $\mathbf{p}_t[i] := P(X_t = s_i)$. Then the forward algorithm can be expressed as

$$\mathbf{p}_t = \mathbf{M} \mathbf{p}_{t-1} \quad \text{i.e.} \quad \begin{bmatrix} P(X_t = s_1) \\ \vdots \\ P(X_t = s_n) \end{bmatrix} = \begin{bmatrix} P(X_t = s_1 | X_{t-1} = s_1) & \cdots & P(X_t = s_1 | X_{t-1} = s_n) \\ \vdots & \ddots & \vdots \\ P(X_t = s_n | X_{t-1} = s_1) & \cdots & P(X_t = s_n | X_{t-1} = s_n) \end{bmatrix} \begin{bmatrix} P(X_{t-1} = s_1) \\ \vdots \\ P(X_{t-1} = s_n) \end{bmatrix}$$

2 Hidden Markov Model (HMM)

2.1 Definitions

When the observed variables are not the states of the Markov process, but are generated by the states, we have a hidden Markov model (HMM).

Sensor Markov Assumption To make it easier, we assume that the current observation depends only on the current state, i.e.,

$$P(\mathbf{E}_t | \{X_t, X_{t-1}, X_{t-2}, \dots, E_{t-1}, E_{t-2}, \dots\}) = P(\mathbf{E}_t | X_t) \quad \text{for any } t$$

Transition Model & Sensor Model In an HMM, we have two models (two CPTs in the corresponding Bayesian network):

- Transition model (same as in Markov processes): $P(X_t | X_{t-1})$
- Sensor model (or observation model): $P(E_t | X_t)$

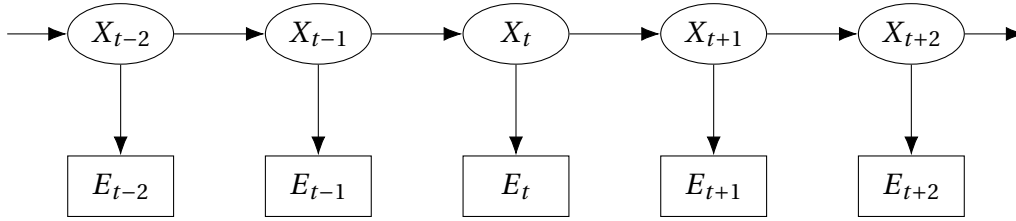


Figure 4: A hidden Markov model (HMM).

2.2 Inference

There are four basic inference tasks in a temporal model:

- **Filtering:** Finding the current state given all evidence up to now, i.e., computing

$$P(X_t | E_1 = e_1, E_2 = e_2, \dots, E_t = e_t) \quad \text{or simply} \quad P(X_t | \mathbf{e}_{1:t})$$

- **Prediction:** Finding the future state given all evidence up to now, i.e., computing

$$P(X_{t+k} | \mathbf{e}_{1:t}) \quad \text{for some } k > 0$$

- **Smoothing:** Finding the past state given all evidence up to now, i.e., computing

$$P(X_k | \mathbf{e}_{1:t}) \quad \text{for some } 0 < k < t$$

- **Most likely explanation:** Finding the most likely sequence of states given all evidence up to now, i.e.,

$$\operatorname{argmax}_{\mathbf{x}_{1:t}} P(\mathbf{X}_{1:t} = \mathbf{x}_{1:t} | \mathbf{e}_{1:t})$$

In addition to the above inference tasks, we also have **learning**: The transition and sensor models, if not yet known, can be learned from observations.

2.2.1 Filtering and Prediction

Let α be the normalization constant that dose not depend on x_t .

$$\begin{aligned} P(X_t = x_t | \mathbf{e}_{1:t}) &= \alpha \cdot P(X_t = x_t, \mathbf{e}_{1:t-1}, e_t) \\ &= \alpha \cdot P(e_t | X_t = x_t, \mathbf{e}_{1:t-1}) \cdot P(X_t = x_t, \mathbf{e}_{1:t-1}) \quad \text{by Bayes' Theorem} \\ &= \alpha \cdot P(e_t | X_t = x_t) \cdot P(X_t = x_t, \mathbf{e}_{1:t-1}) \quad \text{by sensor Markov assumption} \end{aligned}$$

Note that the term $P(X_t = x_t, \mathbf{e}_{1:t-1})$ represents a one-step prediction.

$$\begin{aligned} &= \alpha \cdot P(e_t | X_t = x_t) \cdot \sum_{x_{t-1}} P(X_t = x_t, X_{t-1} = x_{t-1}, \mathbf{e}_{1:t-1}) \\ &= \alpha \cdot P(e_t | X_t = x_t) \cdot \sum_{x_{t-1}} P(X_t = x_t | X_{t-1} = x_{t-1}, \mathbf{e}_{1:t-1}) P(X_{t-1} = x_{t-1}, \mathbf{e}_{1:t-1}) \\ &= \alpha \cdot P(e_t | X_t = x_t) \cdot \sum_{x_{t-1}} P(X_t = x_t | X_{t-1} = x_{t-1}) P(X_{t-1} = x_{t-1}, \mathbf{e}_{1:t-1}) \quad \text{by Markov assumption} \end{aligned}$$

In the above result, the three probability terms on the right-hand side are given by the sensor model, the transition model, and the previous filtering result, respectively. Thus, filtering and prediction can be solved by iterating the above equation.

3 Particle Filtering

4 Kalman filtering

Kalman filters are based on the following linear model:

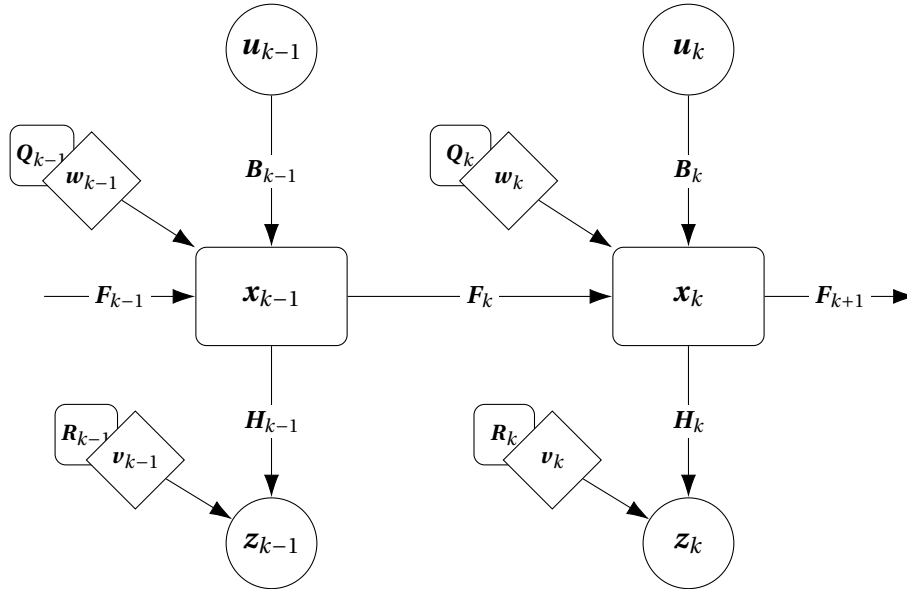


Figure 5: A state-space representation for Kalman Filtering.

Vector $\mathbf{x}_k$ represents the true state of the system at time k , and is evolved from the previous state $\mathbf{x}_{k-1}$ according to

$$\mathbf{x}_k = \mathbf{F}_k \mathbf{x}_{k-1} + \mathbf{B}_k \mathbf{u}_k + \mathbf{w}_k$$

where

- $\mathbf{F}_k$ is the state transition matrix, which models the dynamics of the system.
- $\mathbf{u}_k$ is the control input, which represents any known external influence on the system.
- $\mathbf{B}_k$ is the control input matrix, which models how the control input $\mathbf{u}_k$ affects the state.
- $\mathbf{w}_k$ is a random variable representing the process noise, which is assumed to be Gaussian: $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_k)$

Vector $\mathbf{z}_k$ represents the observation at time k , which is generated from the true state $\mathbf{x}_k$ according to

$$\mathbf{z}_k = \mathbf{H}_k \mathbf{x}_k + \mathbf{v}_k$$

where

- $\mathbf{H}_k$ is the observation matrix, which models how the true state $\mathbf{x}_k$ is mapped to the observation $\mathbf{z}_k$.
- $\mathbf{v}_k$ is a random variable representing the measurement noise, which is assumed to be Gaussian: $\mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_k)$